

Sonifying the Heart: Turning Annotated ECG Recordings into Music

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1 The Project

We are building a system that converts real, cardiologist-annotated electrocardiogram (ECG) recordings into music. The aim is for healthy and pathological cardiac rhythms to sound different because of differences in their musical structure, rather than simply because the underlying ECG traces look different.

Each cardiac cycle contains a clearly detectable landmark: the R peak of the QRS complex. The time between consecutive R peaks, known as the R–R interval, corresponds to the duration between heartbeats. A sequence of R–R intervals therefore naturally defines a rhythm. Our system extracts this rhythmic information from real patient data and uses it as the basis for musical events.

The clinical contrast we focus on is **normal sinus rhythm (NSR)** versus **atrial fibrillation (AFib)**. In NSR, the sinoatrial node fires relatively regularly, so R–R intervals remain close to constant. They vary mainly because of respiratory sinus arrhythmia, which produces a slow and relatively smooth variation of roughly 3–8

Motivation

The main difference between NSR and AFib is not necessarily heart rate. For example, rate-controlled AFib can occur at a normal heart rate of around 70 bpm. Instead, the important difference is the temporal structure of the intervals between beats.

Since rhythm and temporal structure are central components of music, this makes ECG data suitable for sonification. Rather than adding an arbitrary musical representation to the signal, we use the timing of the heartbeats themselves as the basis of the musical rhythm.

2 Exact Goal

The goal is to produce an **open and reproducible parameter-mapping sonification pipeline** that takes recordings from a standard public ECG dataset and generates musically coherent audio in which rhythmic irregularity is perceptually noticeable. The project will also document and justify the mapping decisions used in the system.

2.1 Concrete deliverables

1. A working Python pipeline that takes a PhysioNet record identifier and produces rendered audio end to end, without requiring manual data preparation.
2. A set of paired audio examples containing matched NSR and AFib segments from real annotated records. The segments will be deliberately rate-matched so that differences in tempo do not act as a confounding cue.
3. Supporting figures, including Poincaré plots of R–R intervals for both conditions and plots aligning the raw ECG waveform with the generated musical events.
4. A table of heart-rate-variability statistics (SDNN, RMSSD, and pNN50) for each rendered segment.
5. A written description of the mapping design, including what each musical parameter represents, why the mapping was chosen, and what information is lost or changed as a result.

3 Data

All data comes from PhysioNet and is open access.

- **MIT-BIH Arrhythmia Database (mitdb)** — 48 half-hour two-channel ambulatory recordings from 47 subjects sampled at 360,Hz, with approximately 110,000 beats annotated beat-by-beat and cardiologist-audited.
- **MIT-BIH Atrial Fibrillation Database (afdb)** — 25 long-term records with manually prepared rhythm annotations marking AFIB, atrial flutter, AV junctional rhythm, and normal segments.
- **MIT-BIH Normal Sinus Rhythm Database (nsrdb)** — 18 records with no significant arrhythmia, used as a healthy reference.

3.1 Data-quality considerations

One issue with *afdb* is that the *rhythm* annotations are manually prepared, while the *beat* (.qrs) annotations are machine-generated and unaudited. A false or missed beat detection could introduce artificial irregularity into the R–R series. Since artificial irregularity could sound similar to AFib, this would directly affect the comparison.

We address this in two ways. First, the main NSR/AFib comparison uses the audited beat annotations from *mitdb*. Second, we apply a physiological plausibility filter that discards intervals below 0.3,s or above 2.0,s.

4 Implementation

4.1 Stage 1: Load

Annotations are read using

```
wfdb.rdann(record, 'atr', pn_dir='mitdb')
```

This allows records to be accessed directly from PhysioNet without a manual local download step. Annotation sample indices are converted to seconds using

```
ann.sample / ann.fs
```

4.2 Stage 2: Derive the R–R series

The R–R intervals are computed as

$$rr = np.diff(peak\ times).$$

The resulting intervals are then divided into segments according to their rhythm labels to obtain matched AFib and normal passages. The physiological plausibility filter described above is also applied.

4.3 Stage 3: Quantify

We compute standard heart-rate-variability metrics so that the audio examples are accompanied by numerical measurements:

$$SDNN = sd(rr),$$

$$RMSSD = \sqrt{\frac{1}{N-1} \sum_i (rr_{i+1} - rr_i)^2},$$

and

$$pNN50 = \frac{\#\{|\Delta rr| > 50ms\}}{N-1}.$$

RMSSD is the main metric of interest because it measures beat-to-beat variation. This makes it particularly relevant for distinguishing the irregularity associated with AFib from slower changes such as respiratory sinus arrhythmia.

4.4 Stage 4: Map to music

The main creative component of the project is the mapping from ECG-derived information to musical parameters. We use three mappings.

- **R–R interval → note onset and duration.**

The timing of the notes directly follows the timing of the heartbeats. This is a relatively direct mapping because the R–R intervals already define a temporal rhythm. As a result, irregularity in the cardiac rhythm is preserved in the timing of the musical events.

- **Local R–R deviation → pitch, quantised to a scale.**

We compute a running mean over a short window and calculate the fractional deviation

$$d_i = \frac{rr_i - \overline{rr}_i}{\overline{rr}_i}.$$

When $|d_i|$ falls below a threshold, pitches are drawn from a **minor pentatonic** scale. When the deviation exceeds the threshold, pitches can move to chromatic degrees with octave displacement.

There are two reasons for this design.

First, using a *local* rather than global mean makes the system adaptive to differences in baseline heart rate. For example, both a regular resting rate of 55,bpm and a regular rate of 90,bpm should sound stable, since the purpose of the mapping is to represent irregularity rather than absolute heart rate.

Second, the minor pentatonic scale does not contain semitone intervals, and subsets of the scale tend to remain consonant. Random pitch choices within the scale are therefore less likely to produce strong dissonance. Larger deviations are represented by moving outside this scale, allowing irregularity to affect the harmonic character of the output.

- $|d_i| \rightarrow$ **note velocity.**

The magnitude of the local deviation also controls note velocity. This provides a second perceptual representation of the same information. The purpose is to make larger deviations more noticeable through both pitch and loudness.

4.5 Trade-off

Quantising pitch to a scale reduces precision: a listener cannot recover the exact R–R value from pitch alone. In exchange, the output is more musically coherent and differences may be easier to perceive categorically.

For a music course, we consider this a reasonable trade-off, but it is a deliberate design choice rather than a property of the data itself.

4.6 Stage 5: Render

We generate MIDI using `pretty_midi` and render it using FluidSynth or a DAW. A direct additive-synthesis implementation is also available as a dependency-free fallback.

Timbre, envelope, and pacing are adjusted by ear. The mapping may preserve the relevant information while still producing audio that is unpleasant or difficult to listen to, so rendering choices are also important to the final result.

4.7 Tools

Python — `wfdb` (PhysioNet I/O), `numpy/scipy` (signal processing), `pretty_midi/mido` (MIDI), FluidSynth or a DAW (rendering), and `matplotlib` (figures).

5 Validation Approach

Since we are not conducting listener studies, validation focuses on structural and numerical properties of the mapping.

- **Information preservation.**

Because note onsets correspond directly to R peaks, the inter-onset intervals in the generated MIDI are the original R–R intervals. Therefore, RMSSD computed from the note timings is equal to RMSSD computed from the ECG-derived R–R series by construction. The beat-to-beat timing information used by this metric is preserved by the mapping.

- **Tempo control.**

We select rate-matched segments so that the mean heart rate is comparable across conditions. This helps ensure that audible differences are primarily due to differences in rhythmic structure rather than a simple difference in tempo.

- **Preliminary result.**

In an initial prototype of the mapping and rendering stages, a regular and an irregular series with nearly identical mean heart rates (approximately 70,bpm) produced RMSSD values differing by roughly an order of magnitude: approximately 23,ms versus 217,ms.

This suggests that the mapping responds to irregularity independently of mean heart rate. The prototype used synthetic R–R series with realistic statistics; the final figures will be regenerated using real annotated records.

6 Known limitation

RMSSD alone cannot distinguish AFib from frequent ectopic beats because both can increase beat-to-beat variability.

However, the temporal patterns are different. A premature ventricular contraction typically produces a short–long pattern, consisting of a premature beat followed by a compensatory pause. This pattern has a recognizable structure, whereas AFib produces more broadly irregular timing.

A future extension could use this difference in temporal structure. A mapping sensitive to the short–long asymmetry might represent ectopic beats as a recognizable rhythmic pattern and AFib as a less structured rhythm.